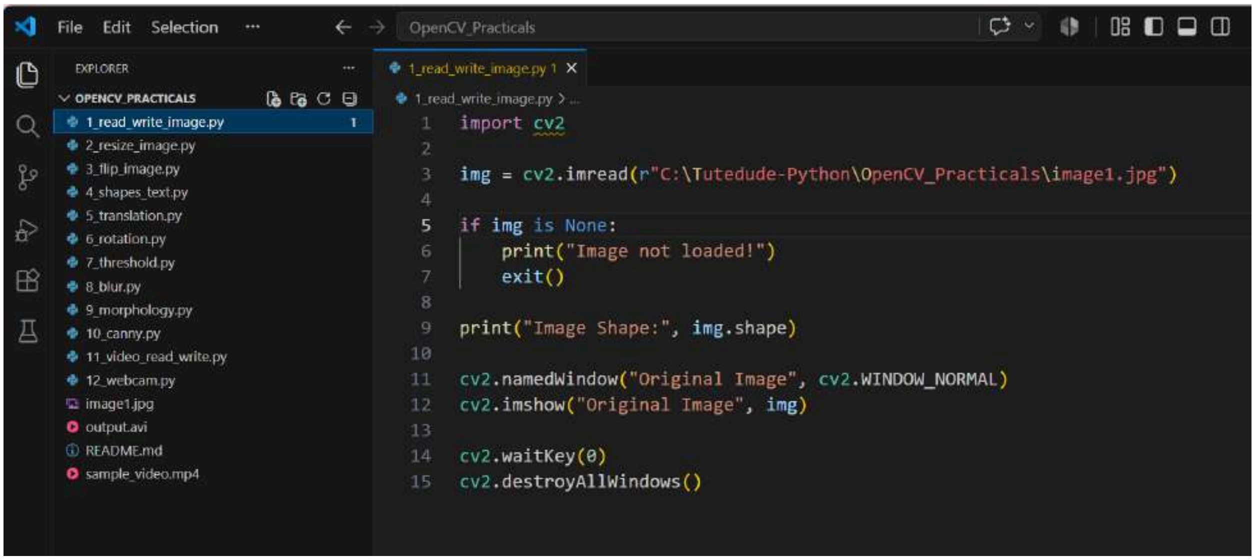
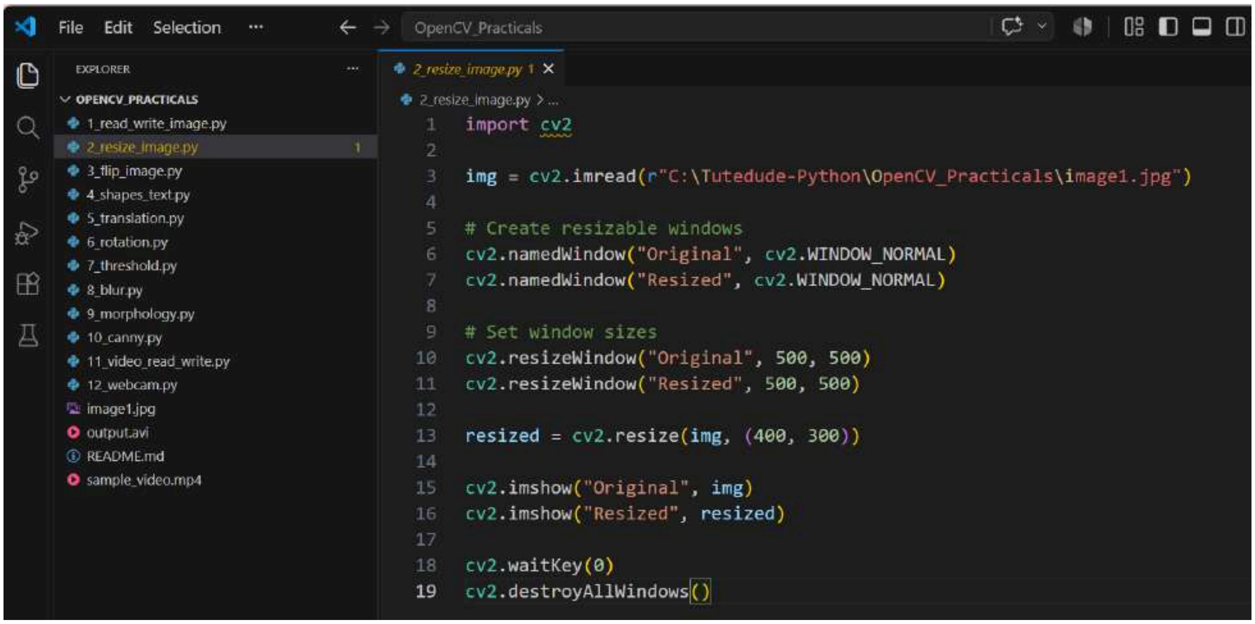
Assignment- OpenCV Practicals

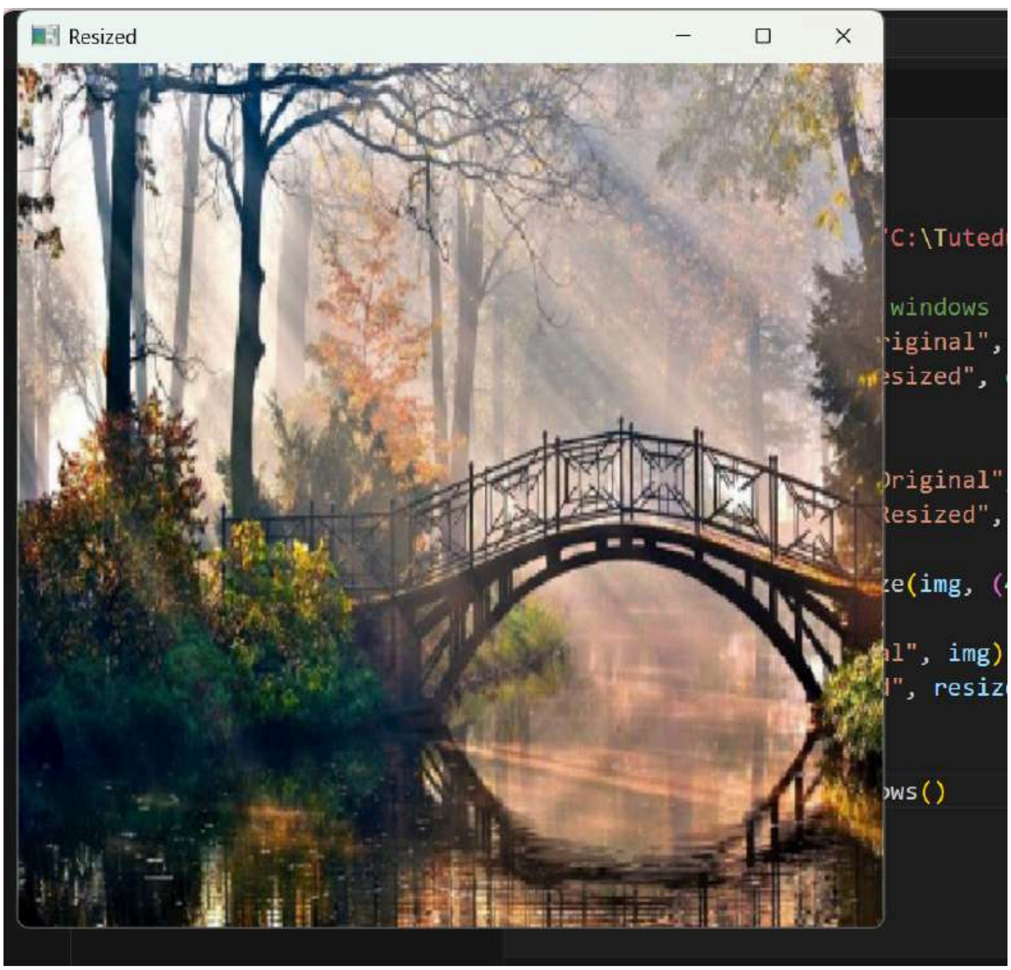


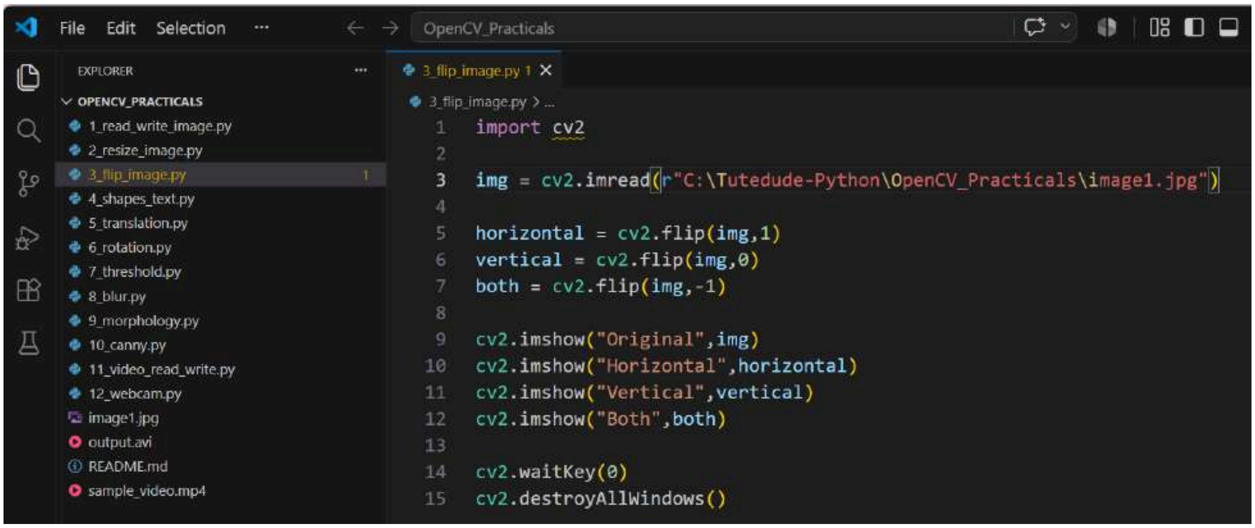
Output:




```python
import cv2

img = cv2.imread(r"C:\Tutedude-Python\OpenCV_Practicals\image1.jpg")

# Create resizable windows
cv2.namedWindow("Original", cv2.WINDOW_NORMAL)
cv2.namedWindow("Resized", cv2.WINDOW_NORMAL)

# Set window sizes
cv2.resizeWindow("Original", 500, 500)
cv2.resizeWindow("Resized", 500, 500)

resized = cv2.resize(img, (400, 300))

cv2.imshow("Original", img)
cv2.imshow("Resized", resized)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:




File Edit Selection ···
OpenCV_Practicals
EXPLORER
OPENCV_PRACTICALS
1_read_write_image.py
2_resize_image.py
3_flip_image.py
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7_threshold.py
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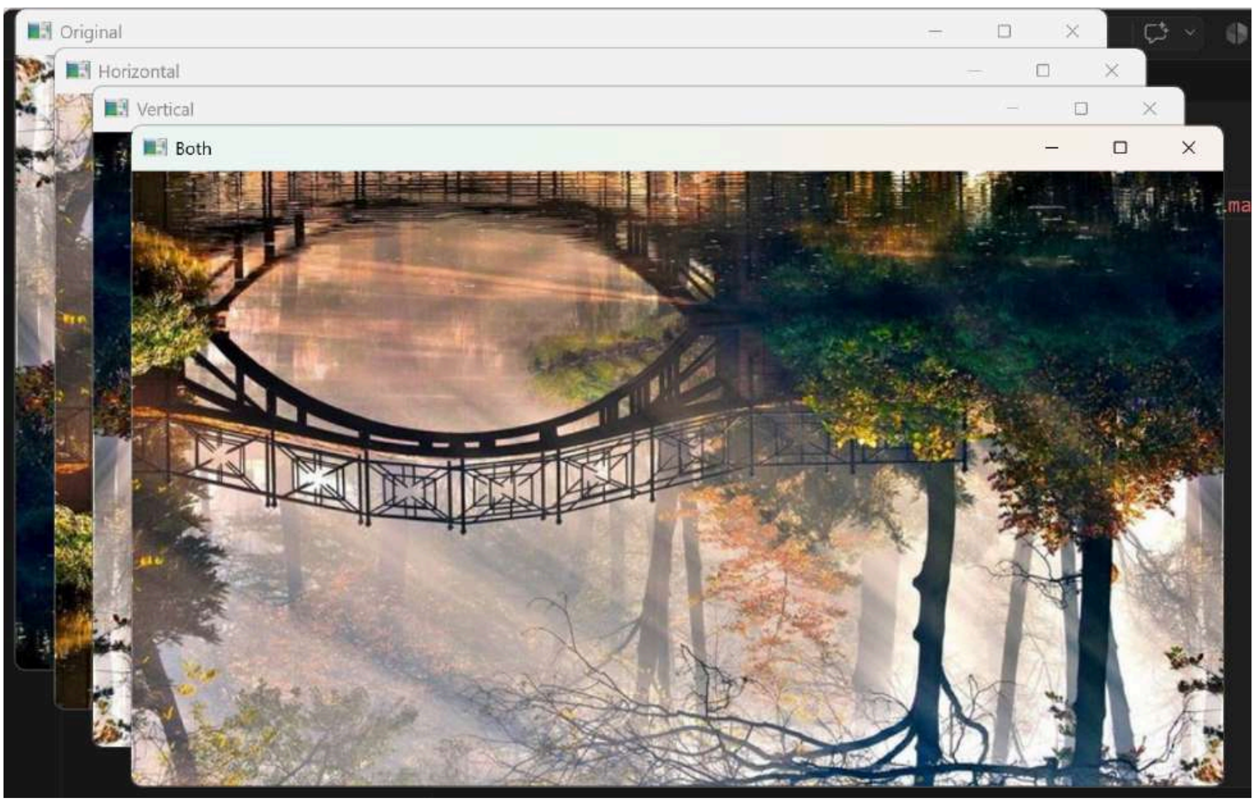
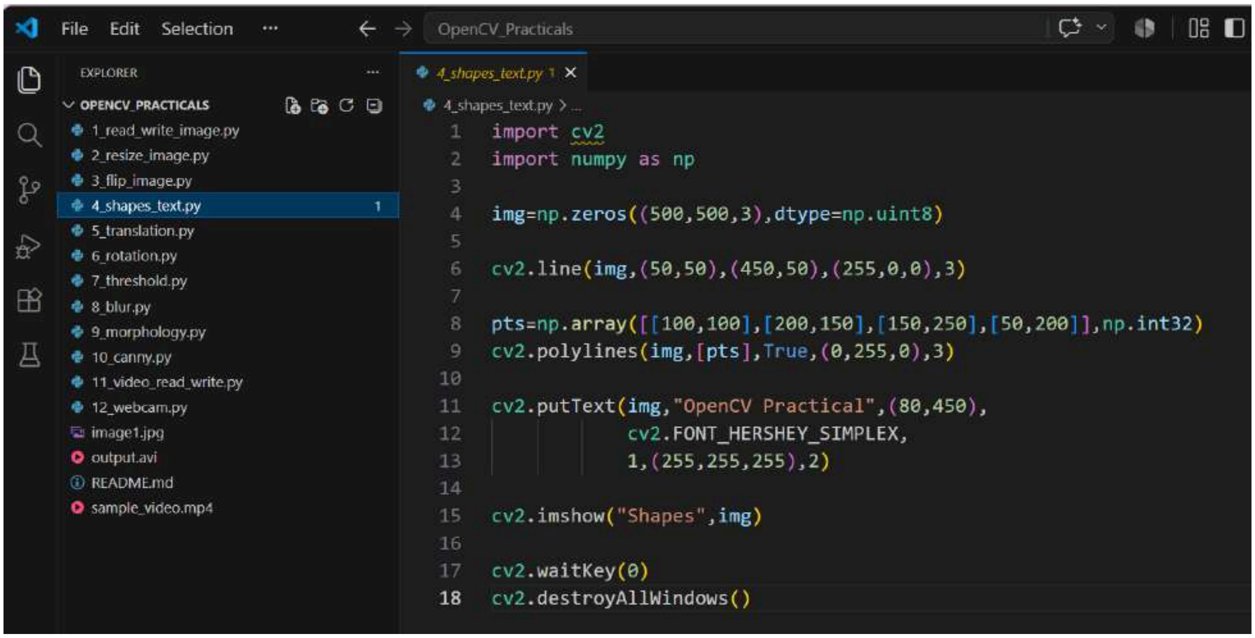
3_flip_image.py 1
3_flip_image.py > ...
import cv2
img = cv2.imread(r"C:\Tutedude-Python\OpenCV_Practicals\image1.jpg")
horizontal = cv2.flip(img,1)
vertical = cv2.flip(img,0)
both = cv2.flip(img,-1)
cv2.imshow("Original",img)
cv2.imshow("Horizontal",horizontal)
cv2.imshow("Vertical",vertical)
cv2.imshow("Both",both)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Horizontal
Vertical
Both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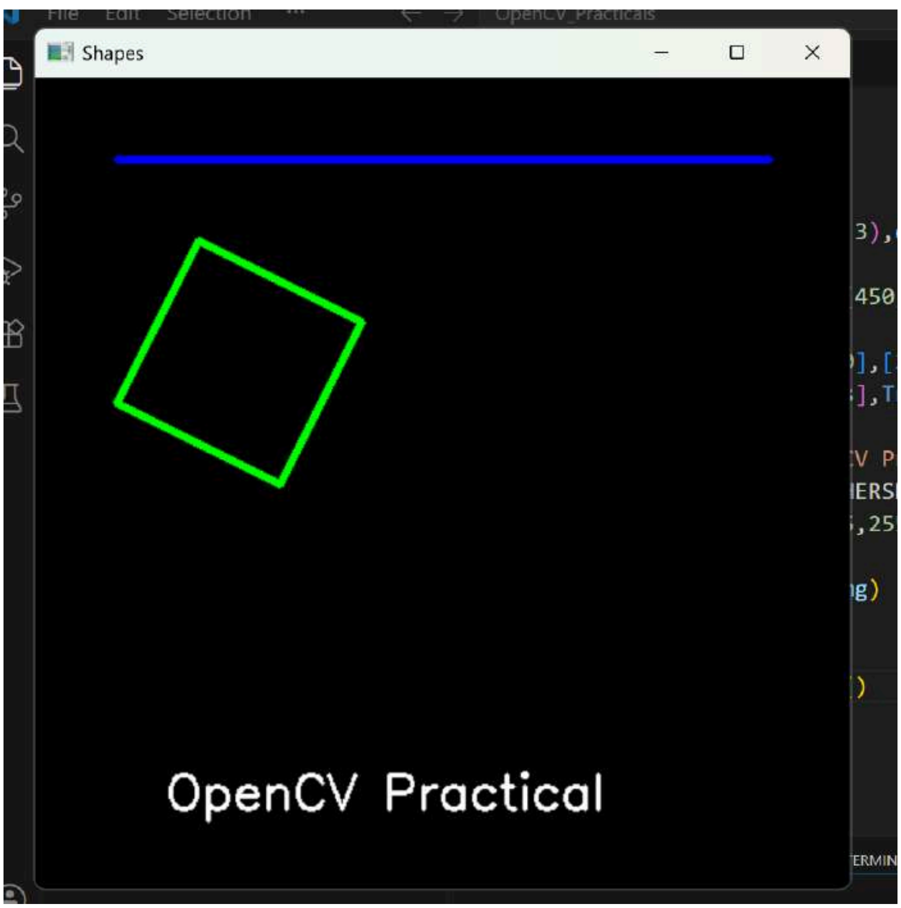
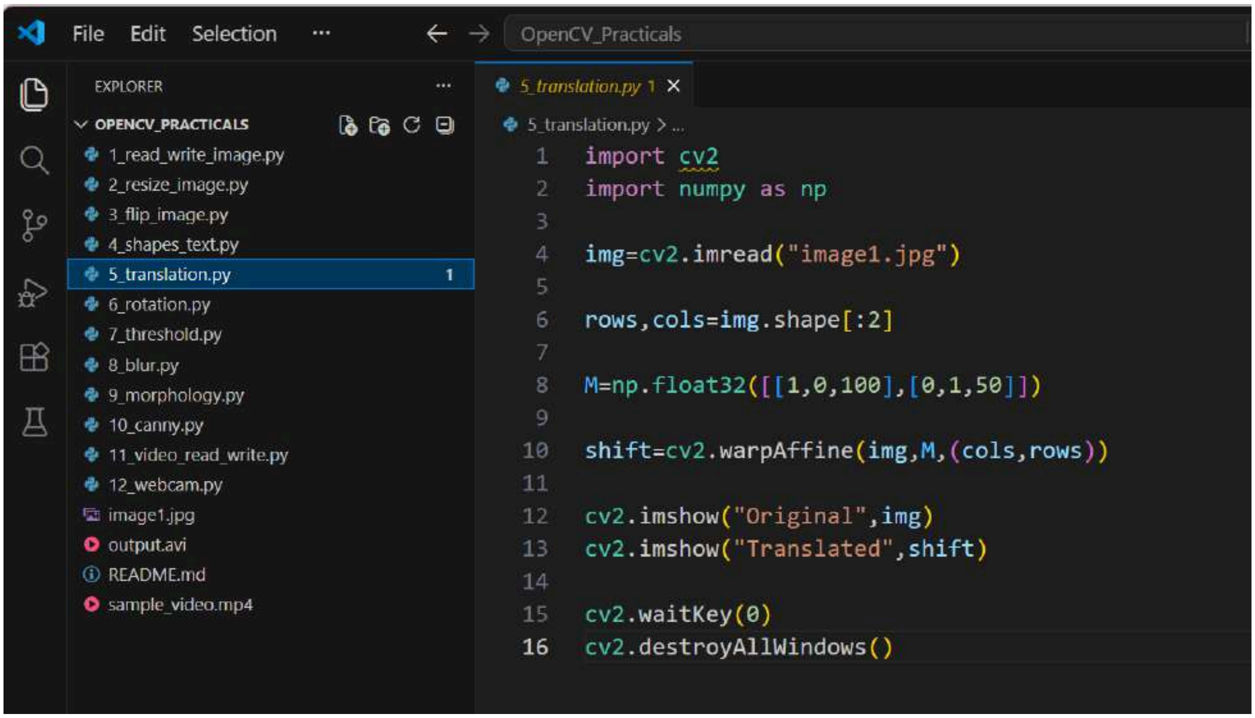
4_shapes_text.py
import cv2
import numpy as np
img=np.zeros((500,500,3),dtype=np.uint8)
cv2.line(img,(50,50),(450,50),(255,0,0),3)
pts=np.array([[100,100],[200,150],[150,250],[50,200]],np.int32)
cv2.polylines(img,[pts],True,(0,255,0),3)
cv2.putText(img,"OpenCV Practical",(80,450),
cv2.FONT_HERSHEY_SIMPLEX,
1,(255,255,255),2)
cv2.imshow("Shapes",img)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Shapes
OpenCV Practical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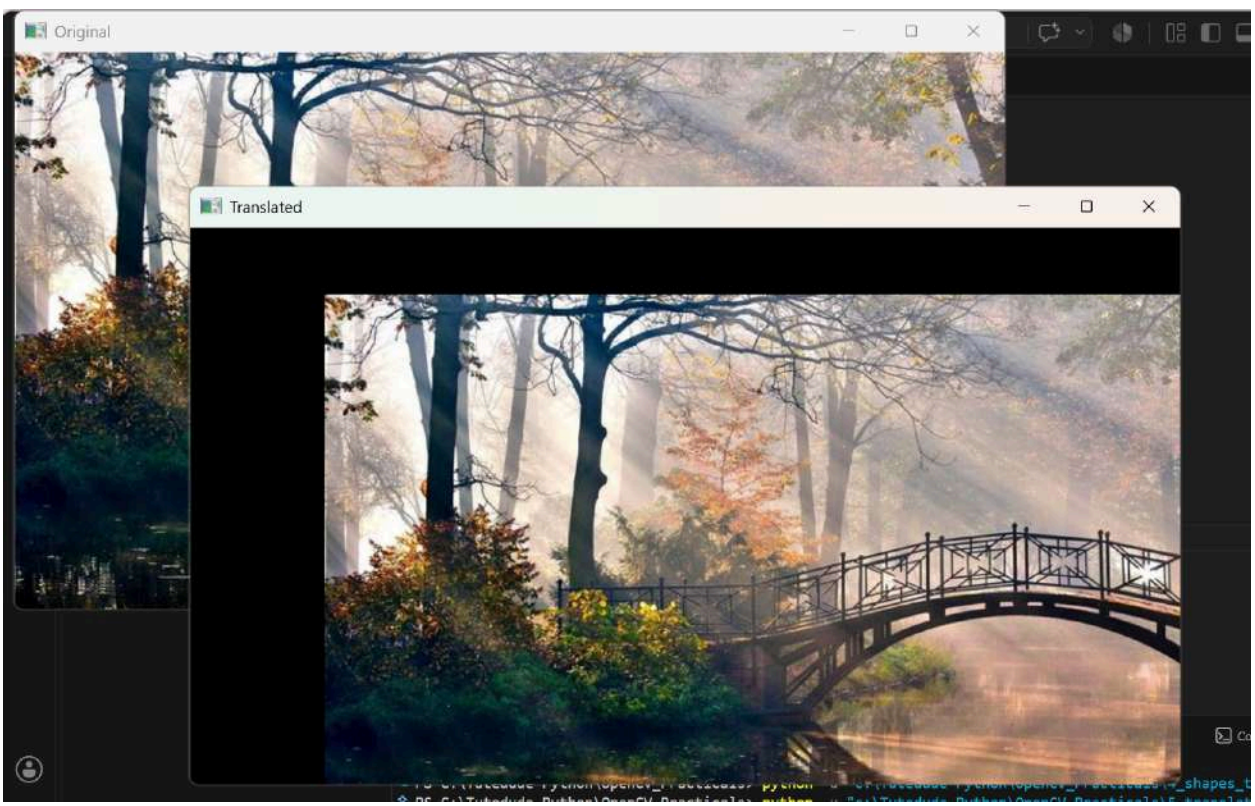
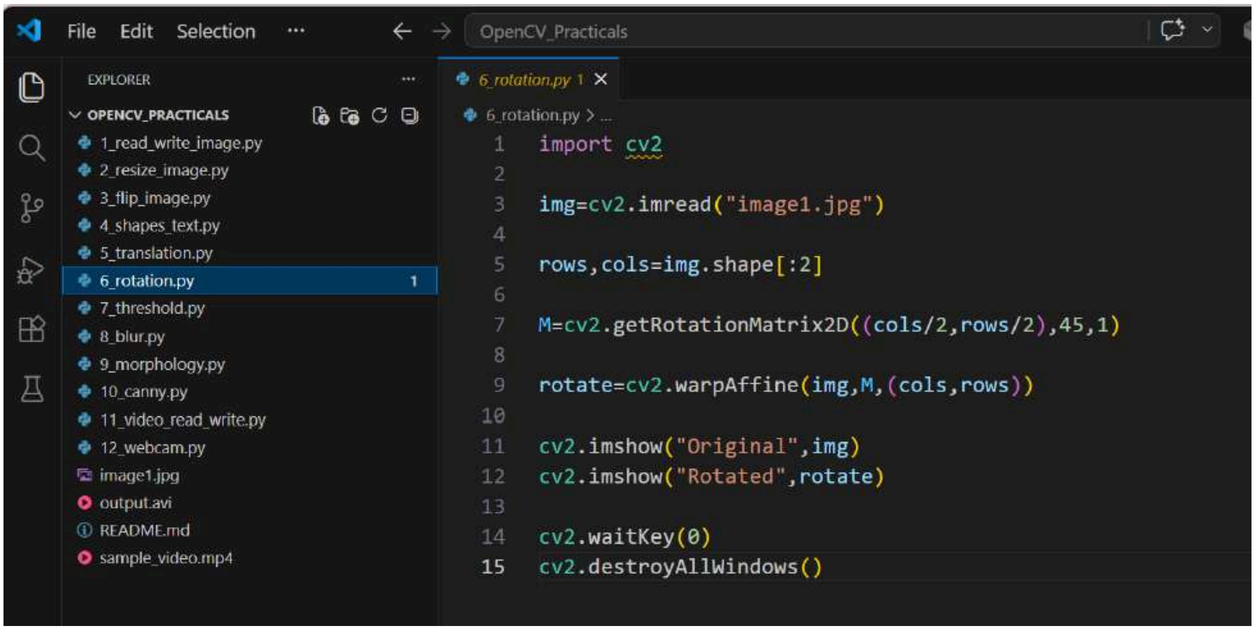
5_translation.py
import cv2
import numpy as np
img=cv2.imread("image1.jpg")
rows,cols=img.shape[:2]
M=np.float32([[1,0,100],[0,1,50]])
shift=cv2.warpAffine(img,M,(cols,rows))
cv2.imshow("Original",img)
cv2.imshow("Translated",shift)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Translated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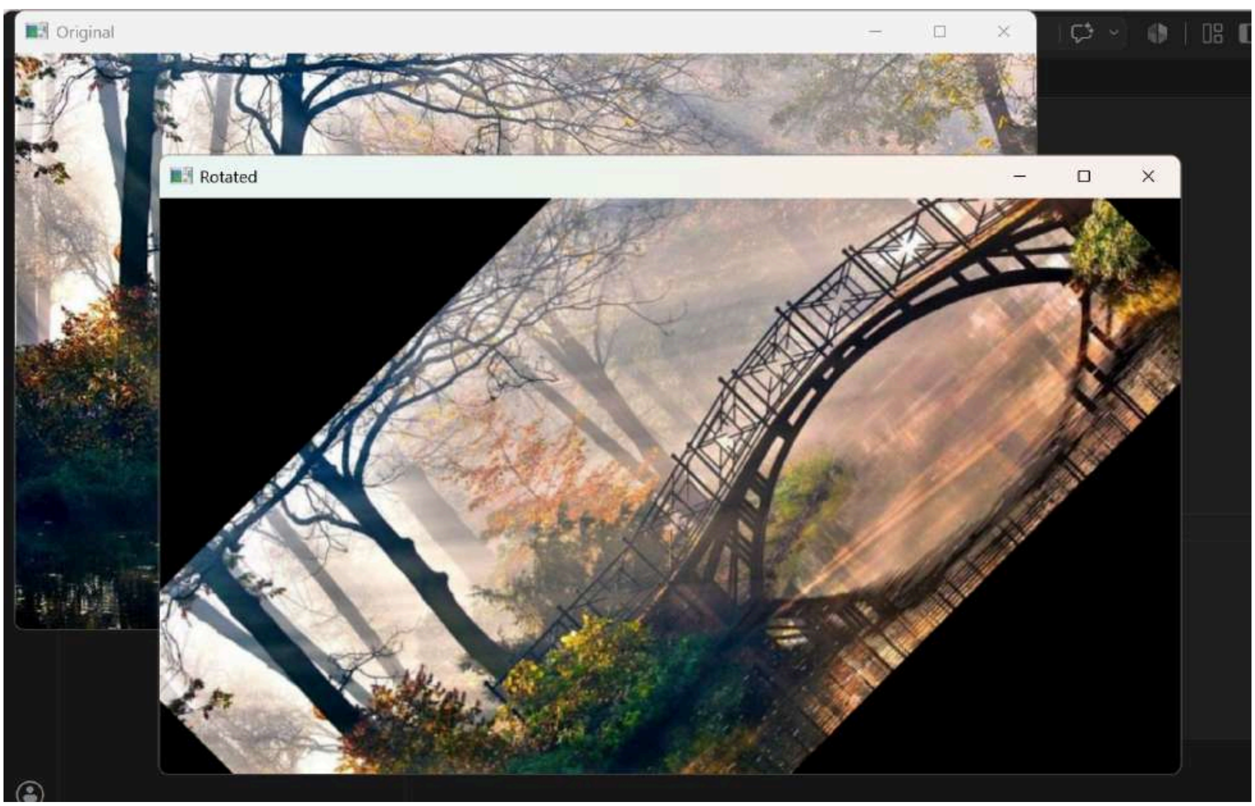
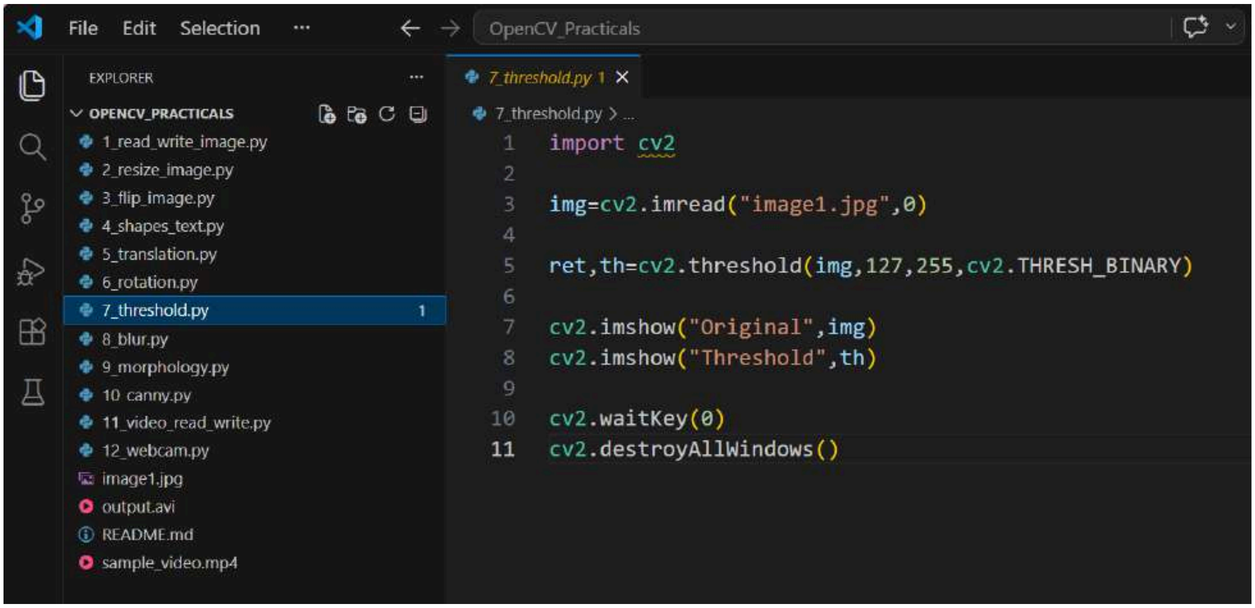
6_rotation.py 1
6_rotation.py > ...
import cv2
img=cv2.imread("image1.jpg")
rows,cols=img.shape[:2]
M=cv2.getRotationMatrix2D((cols/2,rows/2),45,1)
rotate=cv2.warpAffine(img,M,(cols,rows))
cv2.imshow("Original",img)
cv2.imshow("Rotated",rotate)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Rotated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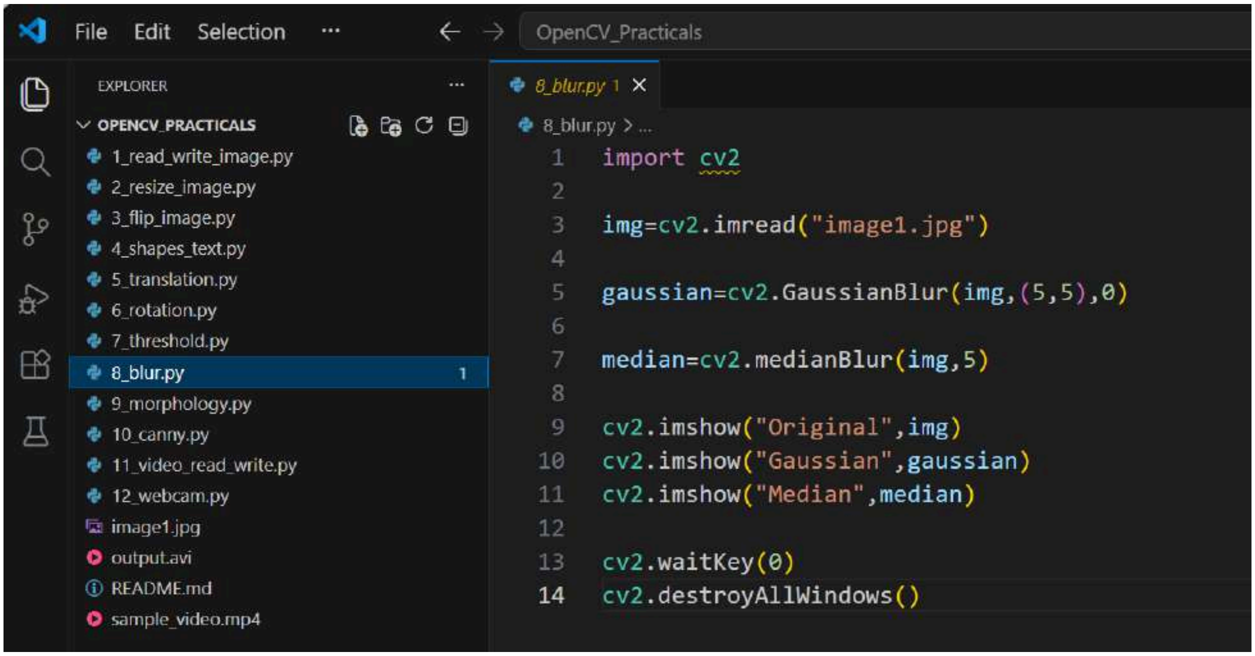
7_threshold.py
import cv2
img=cv2.imread("image1.jpg",0)
ret,th=cv2.threshold(img,127,255,cv2.THRESH_BINARY)
cv2.imshow("Original",img)
cv2.imshow("Threshold",th)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Threshold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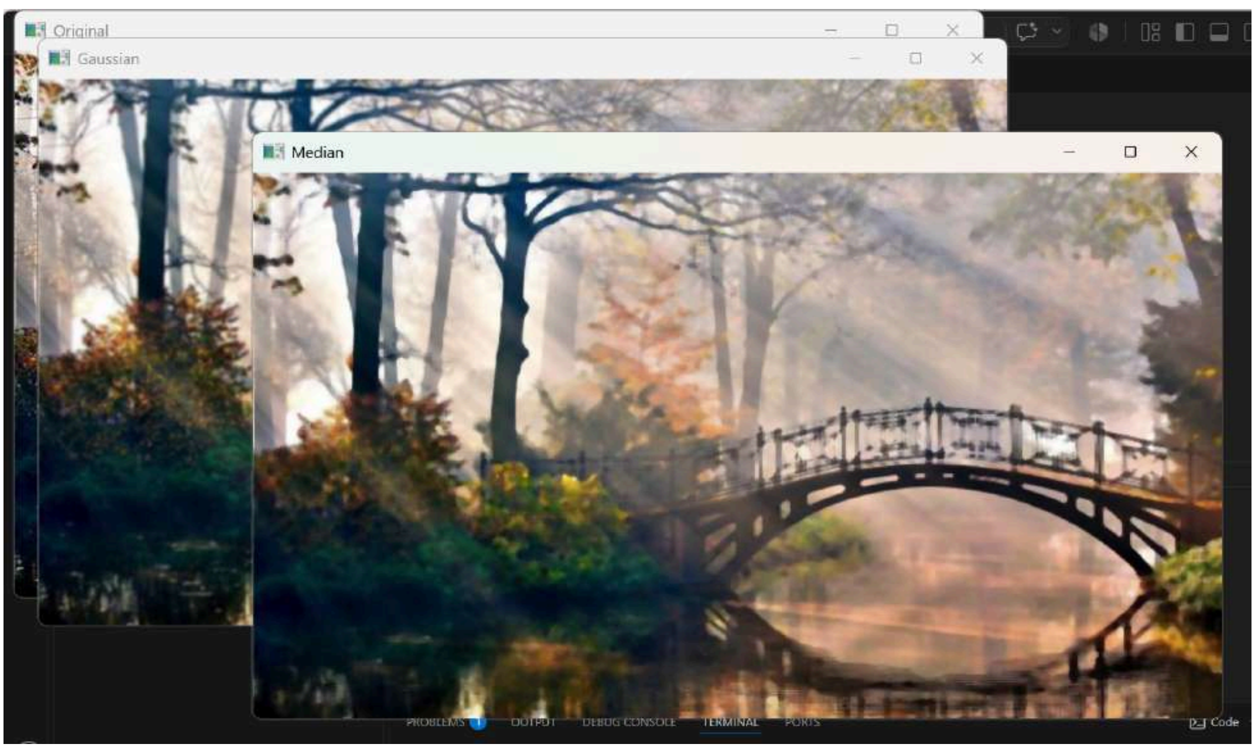
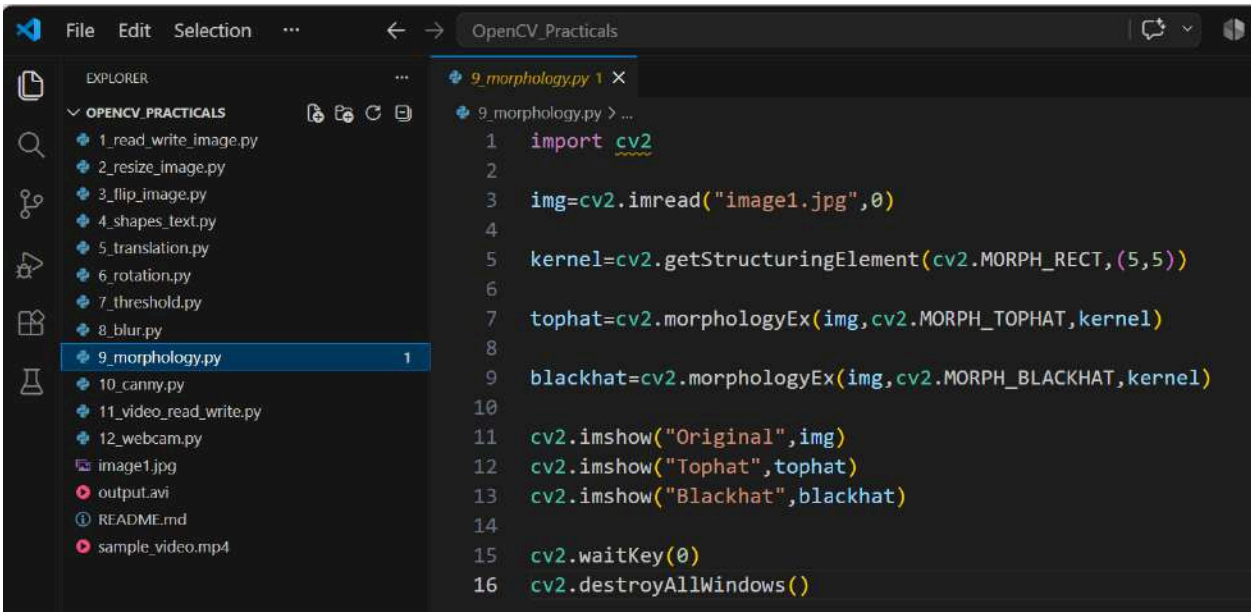
8_blur.py
import cv2
img=cv2.imread("image1.jpg")
gaussian=cv2.GaussianBlur(img,(5,5),0)
median=cv2.medianBlur(img,5)
cv2.imshow("Original",img)
cv2.imshow("Gaussian",gaussian)
cv2.imshow("Median",median)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Gaussian
Median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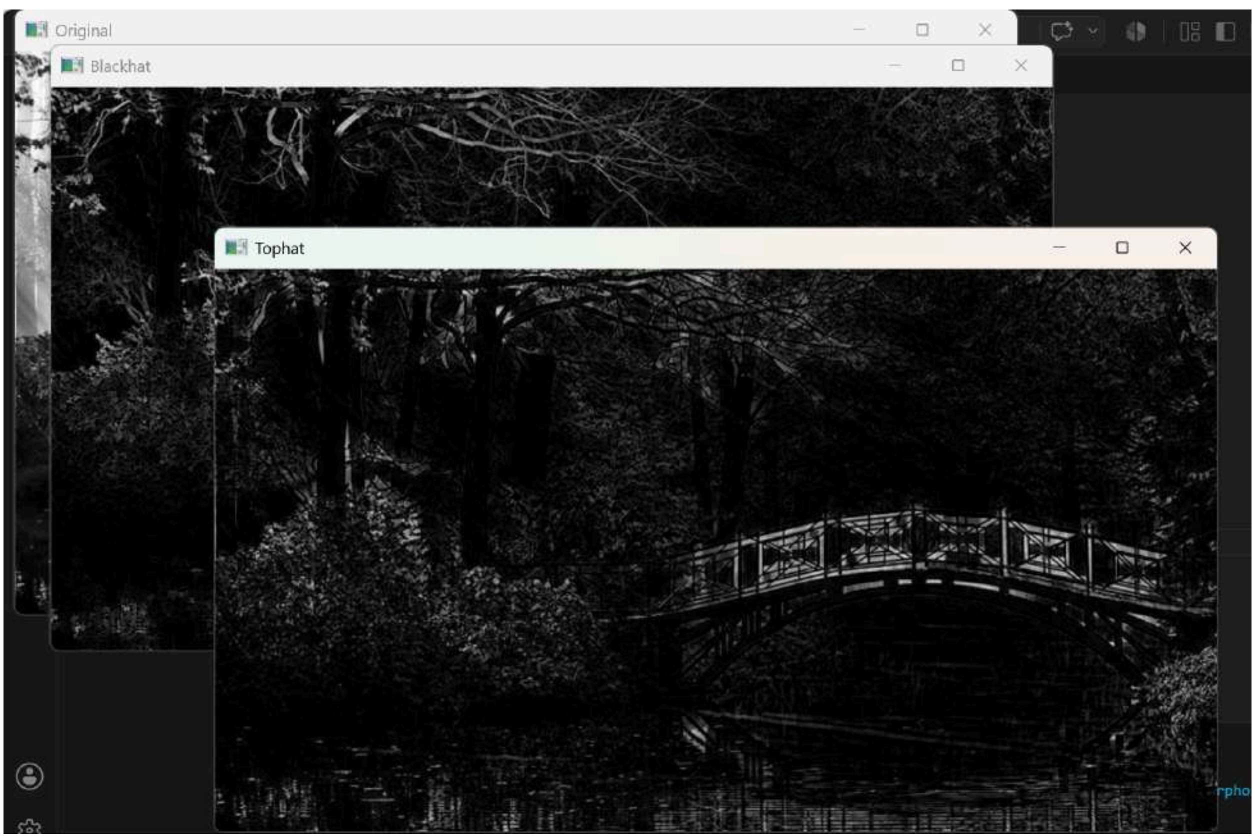
9_morphology.py 1
9_morphology.py > ...
import cv2
img=cv2.imread("image1.jpg",0)
kernel=cv2.getStructuringElement(cv2.MORPH_RECT,(5,5))
tophat=cv2.morphologyEx(img,cv2.MORPH_TOPHAT,kernel)
blackhat=cv2.morphologyEx(img,cv2.MORPH_BLACKHAT,kernel)
cv2.imshow("Original",img)
cv2.imshow("Tophat",tophat)
cv2.imshow("Blackhat",blackhat)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Blackhat
Tophat

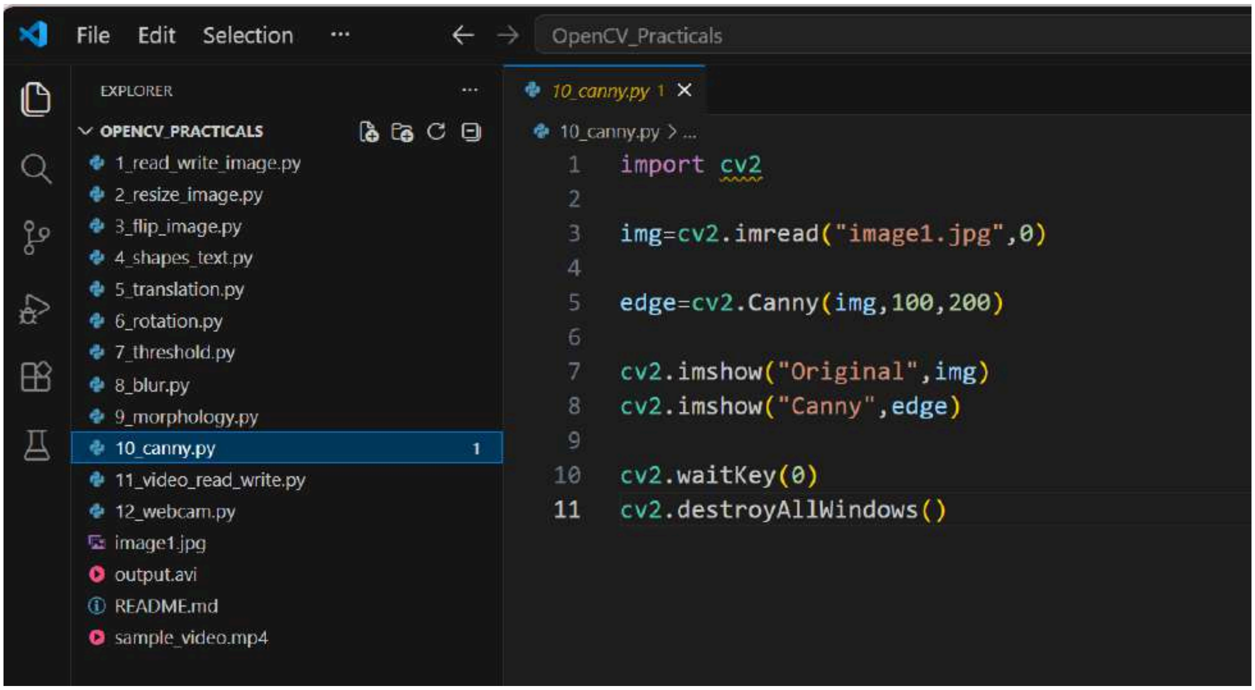

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image1.jpg
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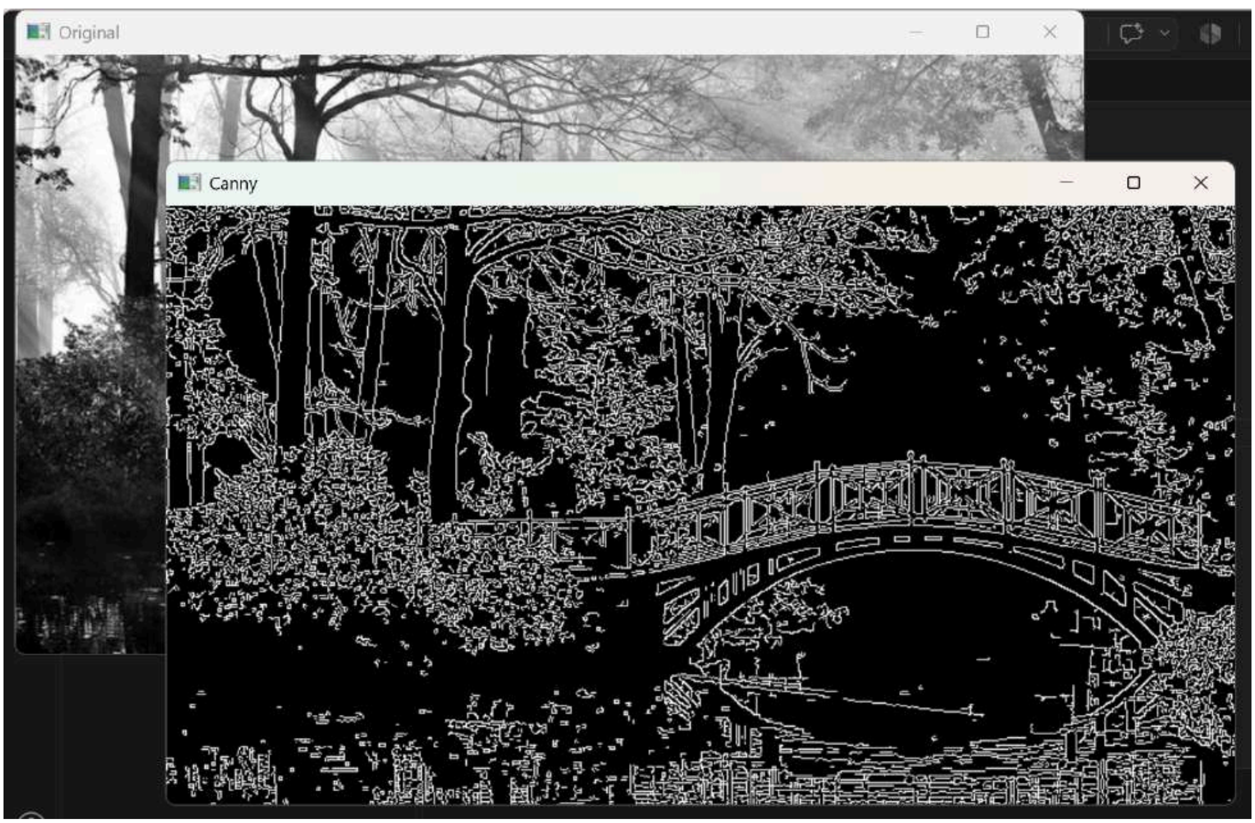
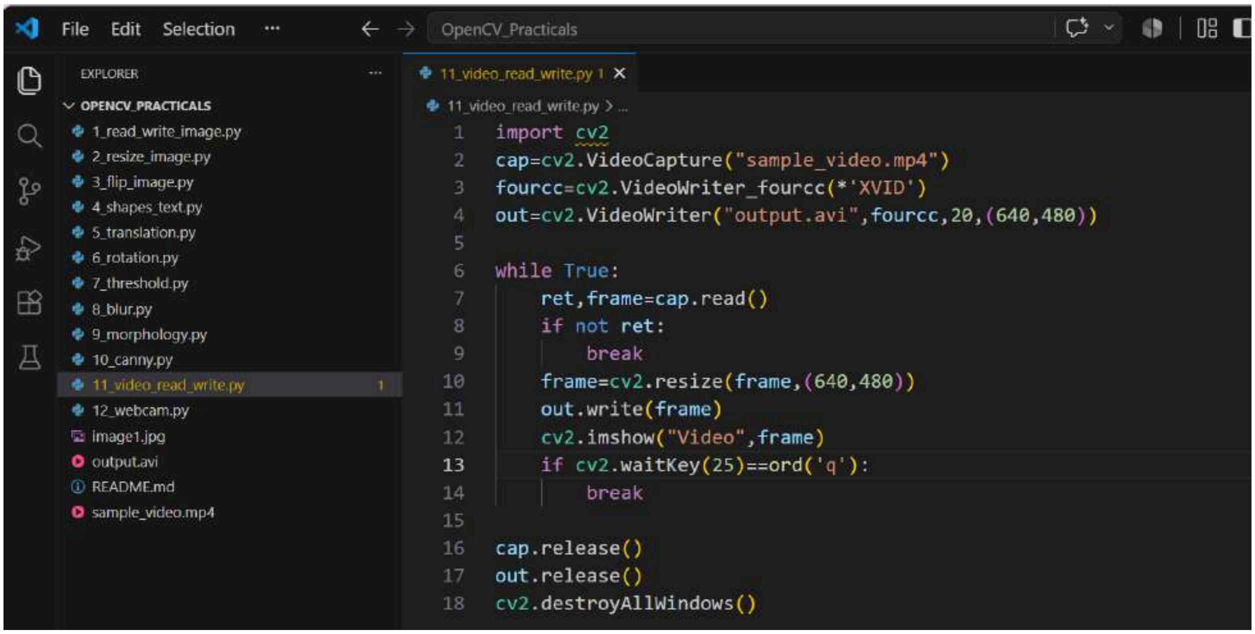
sample_video.mp4
10_canny.py
10_canny.py > ...
import cv2
img=cv2.imread("image1.jpg",0)
edge=cv2.Canny(img,100,200)
cv2.imshow("Original",img)
cv2.imshow("Canny",edge)
cv2.waitKey(0)
cv2.destroyAllWindows()

Output:


Original
Canny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7_threshold.py
8_blur.py
9_morphology.py
10_canny.py
11_video_read_write.py
12_webcam.py
image1.jpg
output.avi
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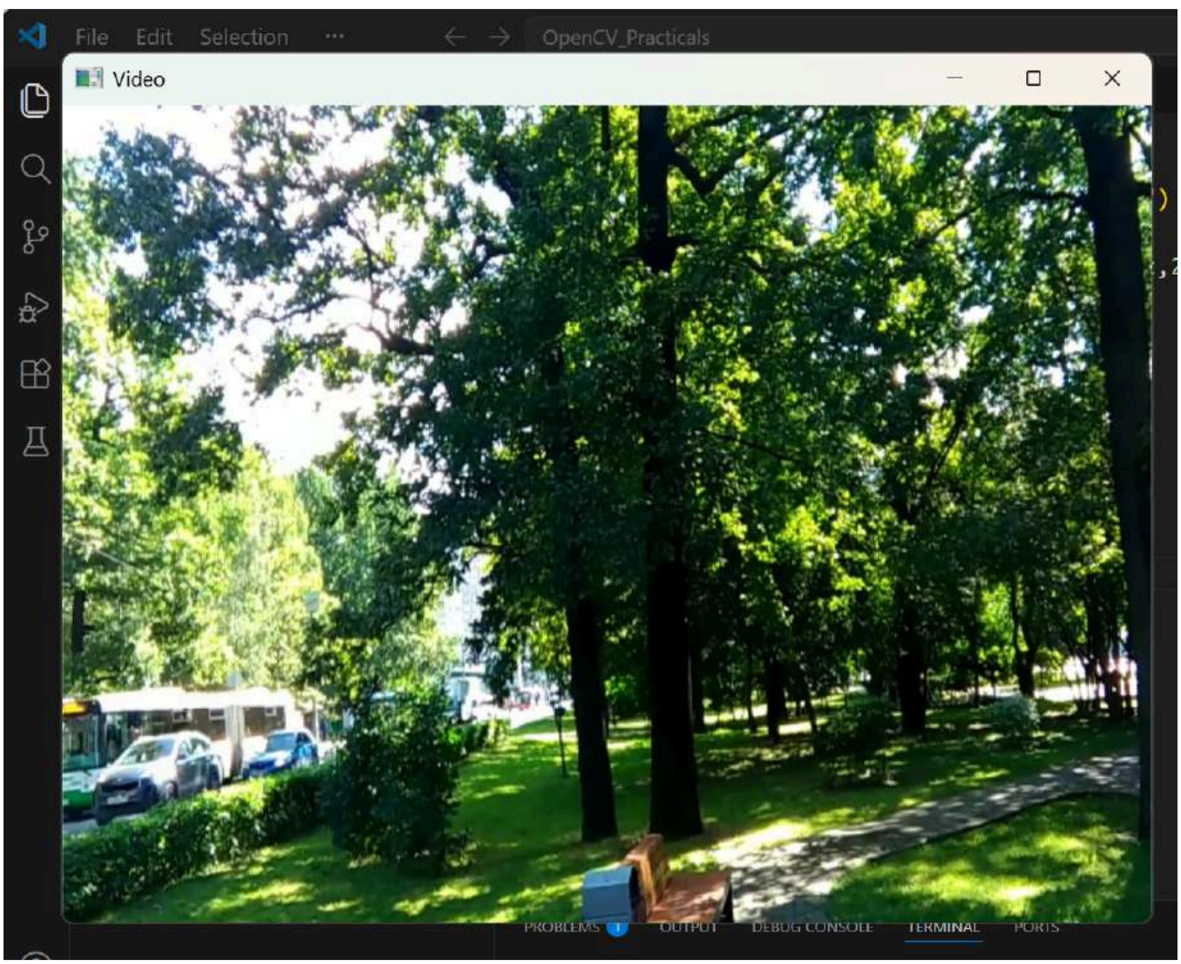
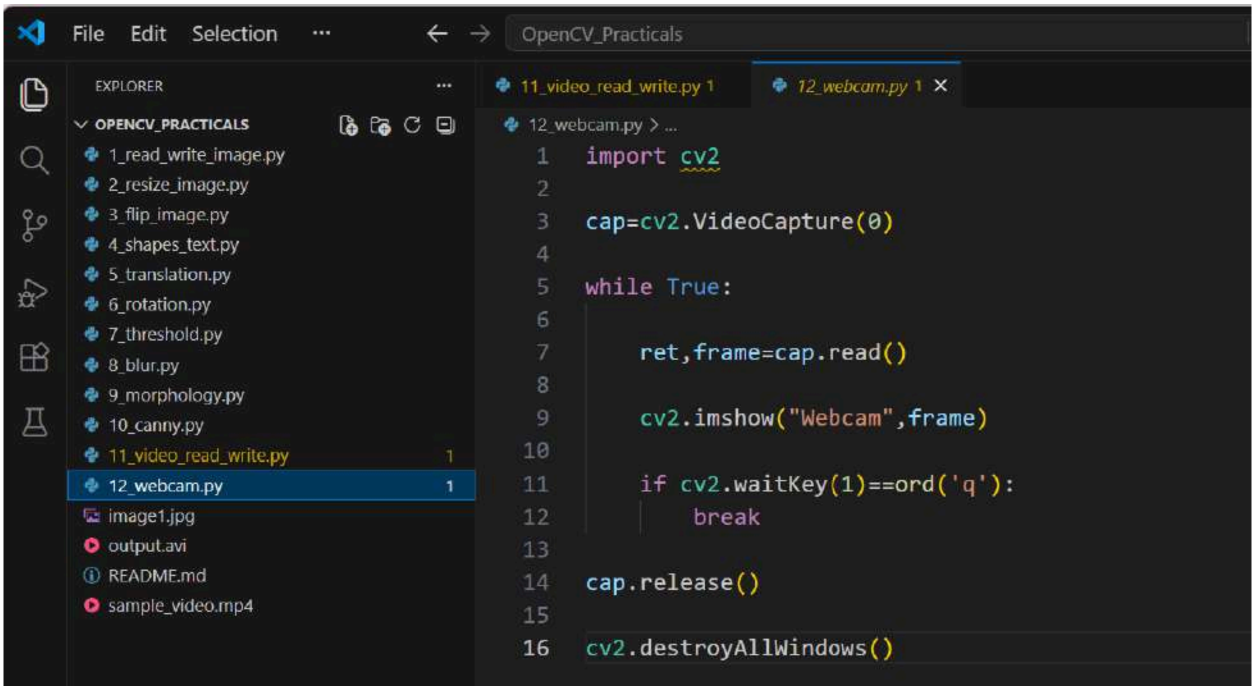
sample_video.mp4
11_video_read_write.py
import cv2
cap=cv2.VideoCapture("sample_video.mp4")
fourcc=cv2.VideoWriter_fourcc(*'XVID')
out=cv2.VideoWriter("output.avi",fourcc,20,(640,480))
while True:
ret,frame=cap.read()
if not ret:
break
frame=cv2.resize(frame,(640,480))
out.write(frame)
cv2.imshow("Video",frame)
if cv2.waitKey(25)==ord('q'):
break
cap.release()
out.release()
cv2.destroyAllWindows()

Output:


File Edit Selection
OpenCV_Practicals
Video
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS


File
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12_webcam.py
image1.jpg
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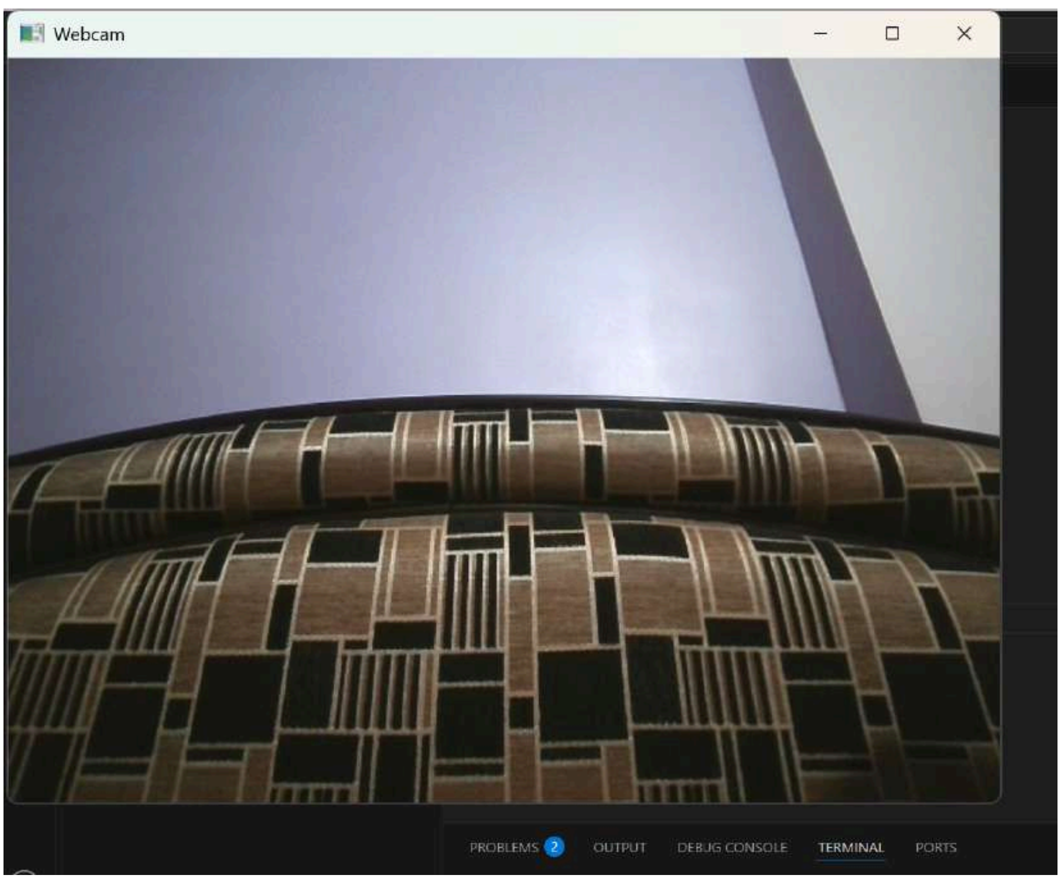
README.md
sample_video.mp4
11_video_read_write.py 1
12_webcam.py 1
12_webcam.py > ...
import cv2
cap=cv2.VideoCapture(0)
while True:
ret,frame=cap.read()
cv2.imshow("Webcam",frame)
if cv2.waitKey(1)==ord('q'):
break
cap.release()
cv2.destroyAllWindows()

Output:


Webcam
PROBLEMS 2
OUTPUT
DEBUG CONSOLE
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